

A consensus void catalogue from 100 ASTRA runs

DESI DR2, LRG and BGS, both caps

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Abstract

ASTRA finds voids with a watershed method. It uses the galaxy data and a random catalogue together. The random catalogue is new in every run, so every run gives a different answer. The distributions are very stable, but single voids are not. This note explains how we build *one* catalogue from 100 runs. The method has six steps. The main product is four FITS files, one per tracer and cap. Every row is a real void from one run. Every row also says in how many runs the region was found. We keep only the regions found in more than half of the runs. We did not choose this limit from our own data. We chose it because it reproduces the size distribution of an independent catalogue, DESIVAST DR1. The shapes then also agree, and we did not use the shapes to choose the limit. So the shape agreement is a test, not a fit.

1 The problem

ASTRA builds a Delaunay tessellation from the galaxies and a random catalogue. Every edge gets a density contrast

$$r = \frac{N_{\text{data}} - N_{\text{rand}}}{N_{\text{data}} + N_{\text{rand}}}, \quad (1)$$

and edges with $r < -0.25$ start the watershed. The random catalogue is made again for every run. So each run is one realization of the same field.

We ran ASTRA 100 times on the same DESI DR2 data. The result is clear and a bit uncomfortable:

- The *distributions* are almost identical. The void counts change by only 1.4%. The median R_{eff} changes by 0.08 Mpc/h.
- The *single voids* do not repeat. Matching one void between two runs works only a little better than chance.

The reason is not that voids are missing. The same empty region is there in every run. But the watershed cuts it in a different way each time. One run says “one big void”. Another run says “five small voids”. We call this **segmentation instability**.

Figure 1 shows the problem and our answer to it. All three panels use the same galaxies in the same slab.

So one run alone cannot tell us which voids are real. A simple union of the runs is also wrong, because it counts the same region many times. And a simple intersection is empty, because no void survives in all runs.

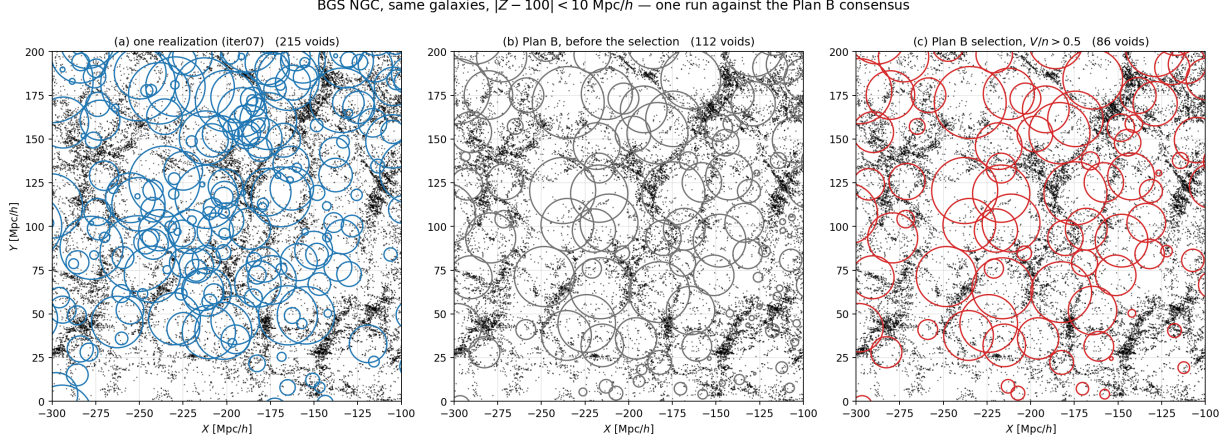


Figure 1: The same galaxies (black dots) in a slab of 20 Mpc/h. **Left:** the voids of one ASTRA run. **Middle:** our catalogue after grouping the 100 runs, but before the last step. **Right:** the final catalogue. The single run has many more circles, and many of them are small pieces. The final catalogue keeps the large empty regions between the filaments.

2 The data

We use the finished catalogues `all.fits` produced by V. Torres, for four cases: LRG and BGS, in the north (NGC) and south (SGC) cap. Each case has 100 runs. We always drop two kinds of voids:

- voids that touch the survey border (**BORDER**);
- voids with an undefined shape. This happens exactly when the void has fewer than four member points. A shape from three points is degenerate, so this is a clean cut and not a loss of a real population.

After these cuts we have 4.1 million pooled voids for LRG NGC and 9.9 million for BGS NGC. BGS voids are about 2.6 times smaller than LRG voids, because BGS is a denser tracer. This is important. Any length in the method must be written in units of R_{eff} , never in Mpc/h. If not, the method will not work for both tracers.

3 The method: six steps

1. **Pool.** Put the voids of all 100 runs in one list. Remember which run each void comes from.
2. **Group, largest first.** Walk through the pooled voids from large to small. When a void j is still free, it starts a new object. Then absorb every pooled void that has more than 50% of its own volume inside the sphere of j . We compute this volume with the exact two-sphere lens formula. This step is deliberately *cross-scale*: the largest description eats the whole nested family below it. So the output is one single-scale catalogue, not a tree.
3. **One vote per run.** A run can put many voids in the same group. We let each run vote only once, with its own largest void in that group. This matters a lot. Inside one group the radii span a factor of about 10. After the vote step the span is only 2.8. We call V the number of runs that vote.
4. **Choose a representative void.** We sort the votes by radius and take the vote in the middle, the medoid. The object then copies *everything* from that one vote: centre, R_{eff} and

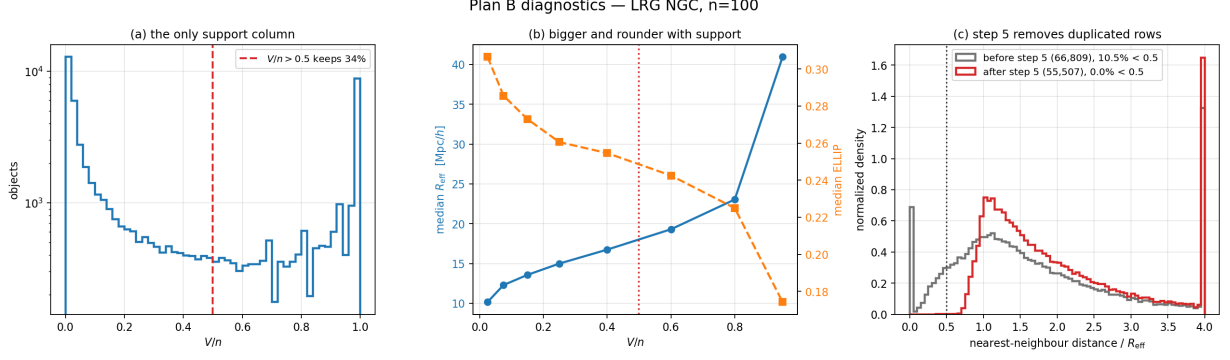


Figure 2: LRG NGC. (a) The support V/n . The red line is our limit. (b) Objects with more support are larger and rounder. (c) The distance to the nearest neighbour, before and after step 5. The copies disappear.

ellipticity. So every row of the catalogue is a void that really exists in one run. We also store which run and which VOID_ID it was.

5. **Prune the catalogue again.** Walk through the representative voids and absorb any that lie more than 50% inside another one. Two representatives can describe the same region, and this step removes the copy.
6. **Keep $V/n > 0.5$.** We keep only the objects that were found in more than half of the 100 runs.

V is the only support number in the catalogue. An earlier version of the method also counted the runs with a void inside the final sphere. We tested both together and they gave the same answer, so we removed the second one.

Step 5 is not cosmetic. Without it, about 17% of the rows describe a region that another row already describes. A simple test shows this: we look at the distance to the nearest other object, in units of R_{eff} . Before step 5, 10.5% of the rows have a neighbour closer than $0.5 R_{\text{eff}}$, and some pairs sit at exactly the same place. After step 5, almost none do. Figure 2 shows this together with the behaviour of V .

4 The four catalogues

catalogue	objects	median R_{eff}	median ELLIP	p16–p84 of R_{eff}
LRG NGC	19 057	32.4	0.195	20.1–52.7
LRG SGC	9 767	31.8	0.195	19.8–52.5
BGS NGC	28 170	15.4	0.182	7.3–27.5
BGS SGC	12 746	13.9	0.189	6.9–25.8

Table 1: The four released catalogues. Radii are in Mpc/ h . The two caps agree very well at fixed tracer. For LRG the ellipticity is the same number in both caps. Nothing in the method was tuned on the caps, so this is a real check of the footprint.

Figure 3 shows the four catalogues together.

The columns are: VOID_ID and SRC_ITER (which void, in which run), the Cartesian position, R_EFF, ELLIP, and V with V/n . We do not give a per-object error bar, and Section 7 explains why.

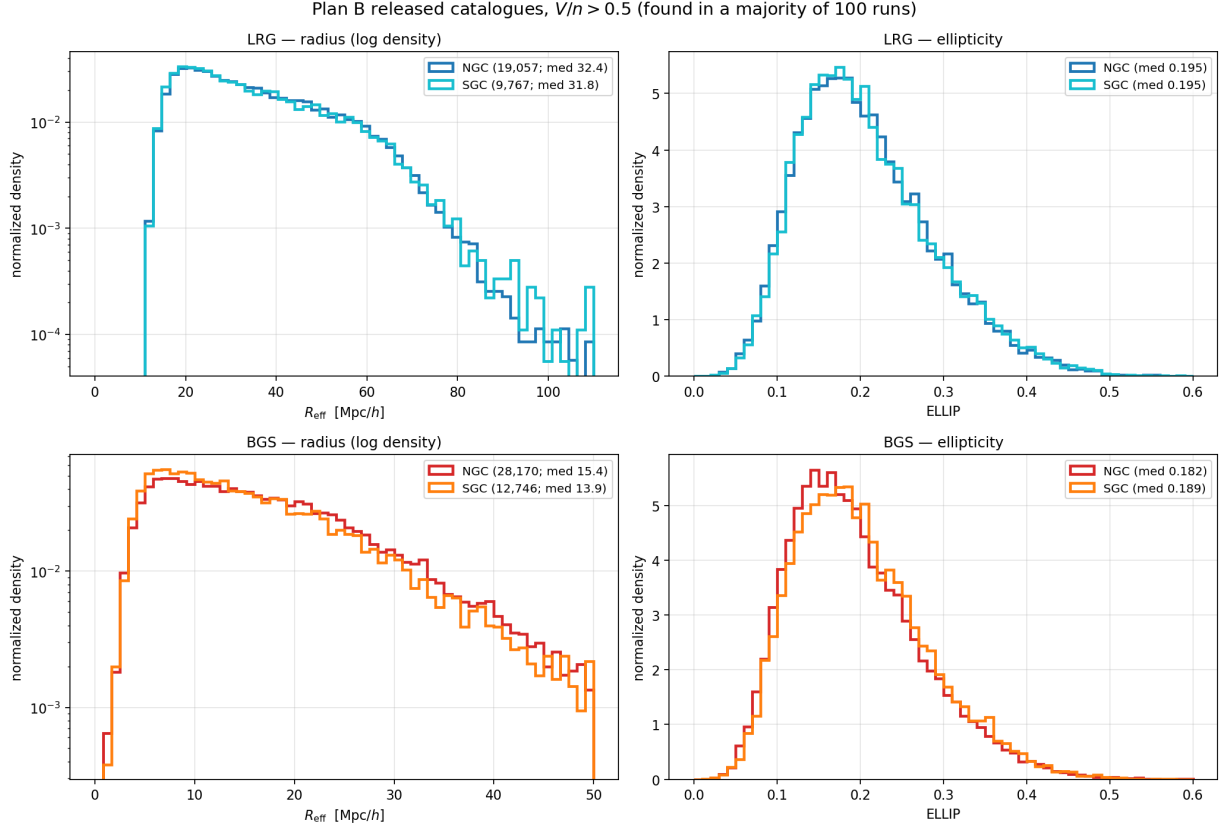


Figure 3: The four released catalogues. Radius uses a logarithmic y axis, because the distribution falls by three orders of magnitude.

5 Where the limit $V/n > 0.5$ comes from

We tried for a long time to choose a support limit from our own data. It does not work. The support histogram has two peaks and a valley, but a random field with the same density gives the same two peaks, so the shape comes from the method and not from the sky. The amount of chance contamination is also flat: it is about 17% whether we cut at 0.5 or at 0.99. A harder cut removes good objects and buys nothing.

So we used an external catalogue instead. DESIVAST DR1 gives BGS voids from the same survey but a different finder. Its V2/ZOBOV version keeps all voids, like ours, while its VIDE and REVOLVER versions remove every void below 10 Mpc/h. ZOBOV is therefore the right partner for our catalogue.

We restrict both samples to $z < 0.24$, because DESIVAST stops there. Then we scan the limit and measure the distance between the two radius distributions with a KS test. Figure 4 shows the result. The best value is 0.525 in the north and 0.550 in the south. The round number 0.5 sits almost exactly on the best value, with KS 0.065 and 0.071. The number of voids per volume also agrees, within 10%.

Two warnings belong with this number. First, we compare *number densities*, never raw counts: our footprint is about 9950 square degrees and theirs is 2450, so the raw counts differ by a factor four for that reason alone. Second, the tracer is not the same. DESIVAST uses a volume-limited sample with a galaxy density of 0.0047 per $(\text{Mpc}/h)^3$. Our runs use a flux-limited sample, which is about 2.9 times denser at the same redshift. So the agreement tells us which selection reproduces their distributions. It does not prove that the two finders agree.

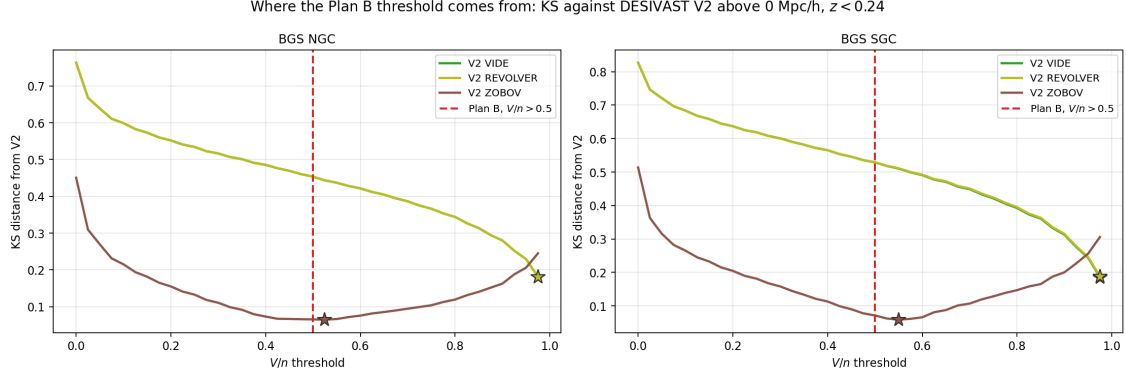


Figure 4: Distance between our radius distribution and the DESIVAST V2 distributions, as a function of the limit. Small is good. ZOBOV has a clear minimum near 0.5. VIDE and REVOLVER never fit well, because they remove all small voids and we do not.

6 The shape test we did not fit

Ellipticity played no part in the choice above. So we can use it as a test.

Figure 5 shows the answer. Before step 6 our shapes do not match ZOBOV (KS 0.228 in the north). After step 6 they do (KS 0.054). The medians are 0.180 against their 0.187 in the north, and 0.189 against 0.201 in the south. A limit chosen only from sizes therefore also reproduces the shapes of another finder.

The right panels show one small disagreement. Our objects are rounder than theirs at every radius, by about 0.02–0.03. Both curves fall with radius in the same way, so this is an offset and not a different relation. The direction agrees with a known problem: ellipticity is too large when a void has few member points, and our rows are single-run measurements. But we have not proved this, and a proper calibration with synthetic voids is still open work.

We also compare the radius distributions directly against all the DESIVAST catalogues in Figure 6.

A note for the DESIVAST team: the axis vectors stored in the V2 files give almost the same value (≈ 0.83) for every void, and they do not correlate with the shape computed from the member galaxies. As shipped they carry no shape information, so we recomputed every shape from the galaxies.

7 What else we tested

These tests were made on an earlier version of the pipeline. They still explain why the final catalogue looks the way it does.

Is the reproducible population real? We shuffled the positions of the pooled voids inside cells and ran everything again. This null field has the same density and footprint but no structure. At high support we found 9351 real objects and only 1658 in the null. That is an excess of 82%, so about 18% of that class comes from chance. The null also produced the same two-peak support histogram, which is why we stopped reading that shape as evidence.

Can we give an error bar per object? We split the 100 runs into two *disjoint* halves of 50, built a full catalogue from each, and matched them. This gives two independent measurements of the same region. The spread of the votes is *not* an error bar: it is about 2.6 times larger than the real scatter. We also expected the noise to fall like $1/\sqrt{K}$ when we add runs. It does not.

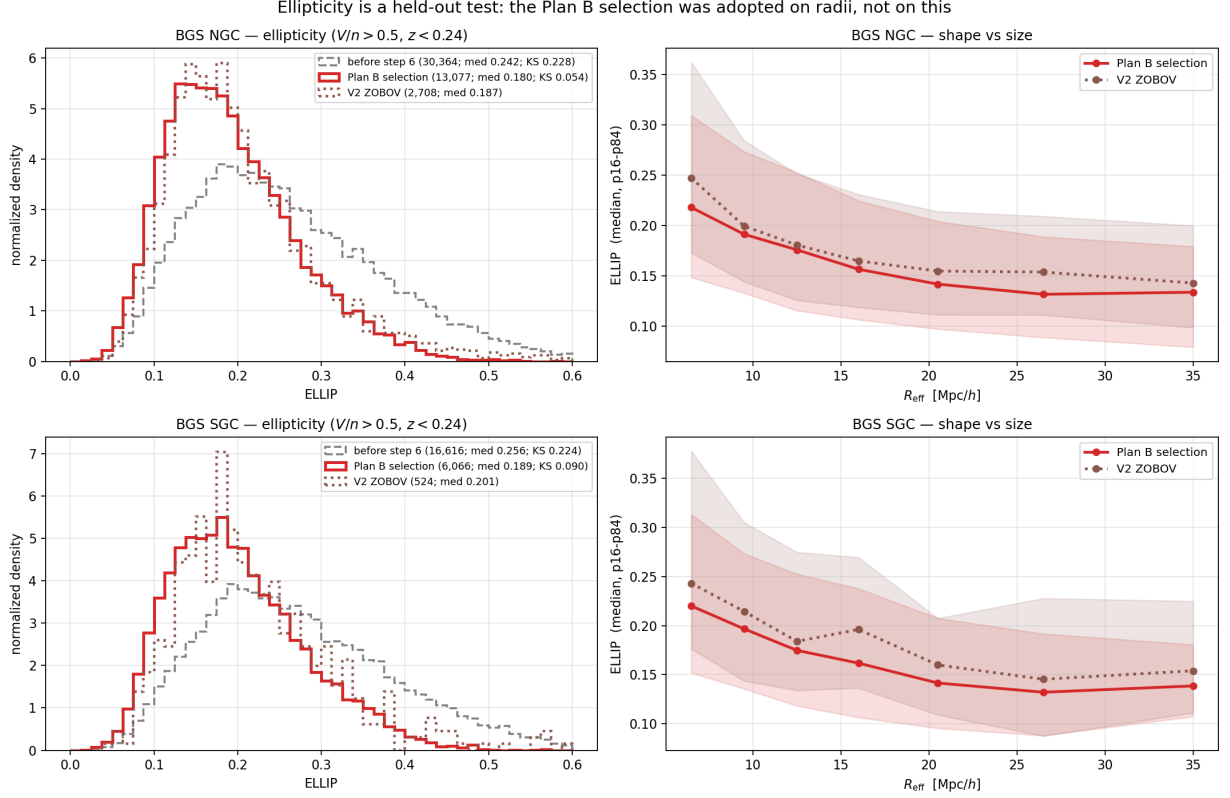


Figure 5: Ellipticity, before and after step 6, against V2 ZOBOV. The right panels show shape against size, where neither catalogue was tuned. The two fall on the same slope, with a small offset.

Going from $K = 25$ to $K = 50$ improves the radius by a factor 1.13, not 1.41. There is a floor near $2.8 \text{ Mpc}/h$.

For this reason the released catalogue has **no error-bar column**. We prefer to give none than to give one that a user would read as a 1σ uncertainty. The same test did give one useful result: the relative scatter of the radius falls from about 0.17 at low support to about 0.07 at high support, and the same three numbers appear in all four catalogues. So support predicts precision, even if it does not predict purity.

Figure 7 shows this test.

Are the low-support objects pieces of larger voids? This was our best guess for what they are. We built the test: for a small object and one *specific* large object, count the runs in which the small one lies inside the large one. The answer is no. The class we looked for is only about 1–2% of the catalogue, and its excess over the null is not stable between the two caps. So the many low-support objects are not a hidden hierarchy. They are single-run segmentations, and step 6 removes them.

8 How to use the catalogue

Please do. Use V/n , never the absolute number of runs. Say which catalogue and which n you used. Quote counts together with the systematic error, which is different for each tracer.

Please do not. Do not read the two peaks of the support histogram as two populations: a random field gives the same shape. Do not raise the support limit to gain purity: the contamination

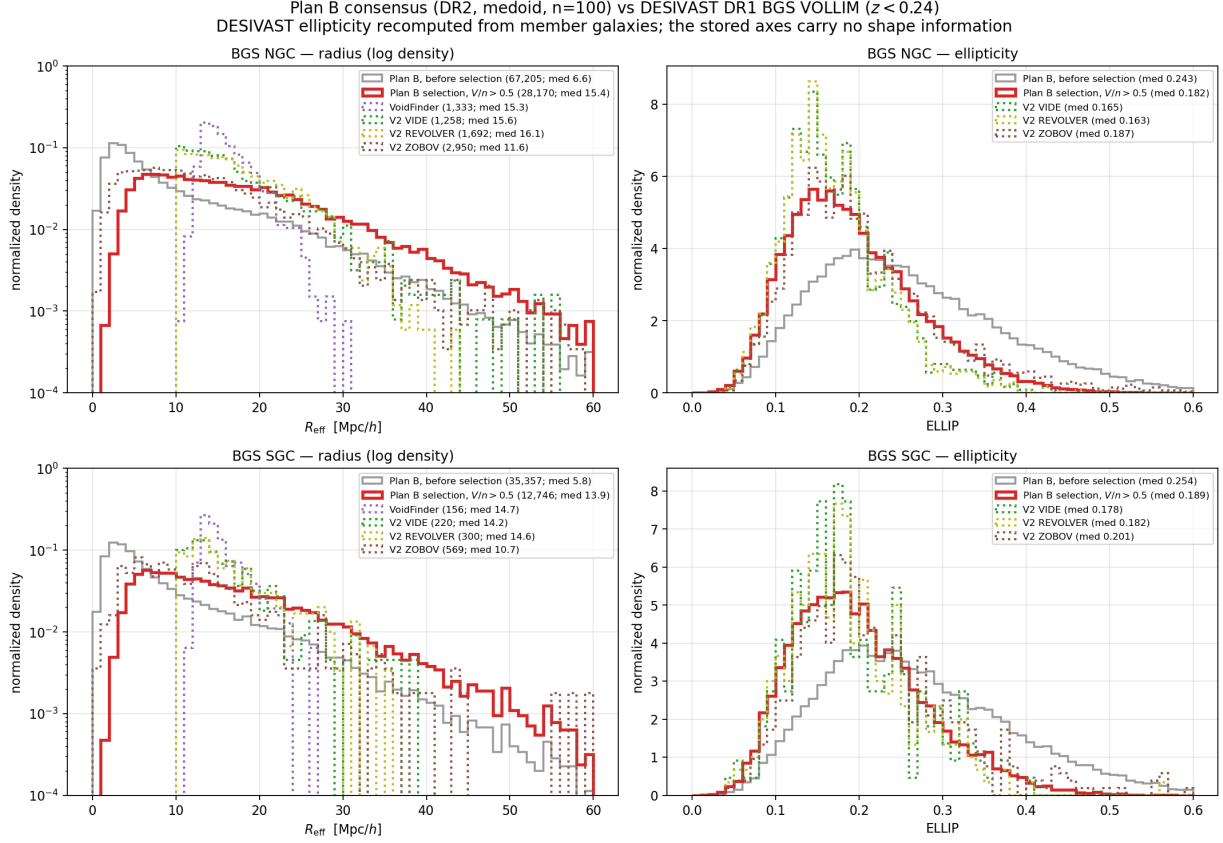


Figure 6: Our BGS catalogue against DESIVAST DR1, before and after step 6. The y axis is logarithmic. Ellipticity for DESIVAST is recomputed from the member galaxies with our own estimator.

is flat. Do not say that BGS voids are more reproducible than LRG voids: we still need a test at matched density. Do not read the fall of ellipticity with support as a better shape measurement: it is a change of population, because objects with high support are large, and large voids are rounder. Do not use the limit $V/n > 0.5$ on LRG as if it were tested there: DR1 has no LRG void catalogue, so for LRG the limit is an extrapolation from BGS.

9 Summary

We turned 100 unstable segmentations into one catalogue of stable objects. Every row is a real void from one run, and every row carries the number of runs that agree.

The step that makes the result defensible is the last one, and it does not come from our own data. We chose the limit so that our size distribution matches an independent catalogue. The shapes then matched as well, and we did not use them to choose. That is the strongest external evidence we have that these objects are real regions and not an artefact of our own grouping.

Two things stay open. The shape of a void is biased when the void has few members, and no number of runs can fix that; it needs synthetic voids. And a comparison between BGS and LRG needs a test at matched galaxy density.

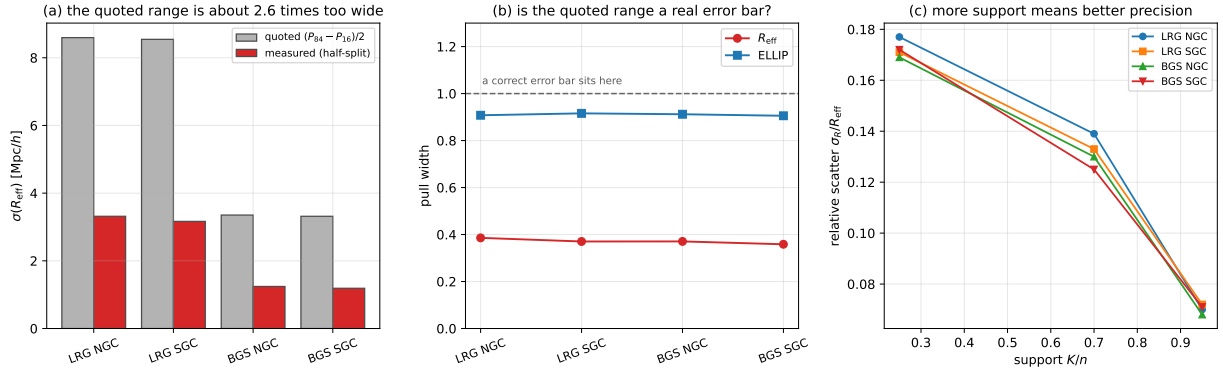


Figure 7: The half-split test. **(a)** The quoted range against the scatter we really measure. **(b)** A correct error bar would give 1. **(c)** The relative scatter falls with support, in the same way for all four catalogues.